Earth and Planetary Science Letters, 36 (1977) 269–279
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A PALAEOMAGNETIC EXPERIMENT ON CRUSTAL UPLIFT IN WEST GREENLAND

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Received February 23, 1977
Revised version received June 3, 1977

In the Nordre Strømfjord shear zone of central West Greenland it is believed that the gneisses exposed at the coast reached a maximum depth of burial about 10 km deeper than those now adjacent to the inland ice. This theory was tested palaeomagnetically. Further sampling up the mountain Kællingehætten was carried out to ascertain changes of direction of magnetization with height.

The magnetic susceptibility ellipsoids were determined for each specimen and were used to correct their directions of magnetization for the effects of anisotropy. Complicated but compact polar wander paths were derived for both the fjord and mountain traverses. The path for the fjord is consistent with the geological model but does not support it. This inconclusive result may be the consequence of a low rate of polar wander. The mean palaeomagnetic poles are 27.7°N, 276.6° E (dp = 13.8°, dm =17.3°) for Nordre Strømfjord and 25.3°N, 300.4°E (dp = 11.5°, dm = 14.1°) for Kællingehætten.

1. Objectives

Geological investigations (1966–1969) in the Agto-Nordre Strømfjord area led to the recognition of a major ductile shear zone [1] that can be traced from the coast to the inland ice (Fig. 1). Compositional layering and schistosity of the rocks within the shear zone strike constantly NE-SW, and in general dip steeply; a subhorizontal linear fabric is well developed in some lithologies. Large-scale dome-and-basin-like structures prevail outside the shear zone.

The part of the shear zone at the coast, mapped during this first project, turned out to be wedge-shaped, narrowing upwards. A significant decrease in width from the coast to the ice (from approximately 15 km to 7 km) and a concomitant decrease in metamorphic grade from within granulite facies to within amphibolite facies could be explained, assuming the wedge shape of the shear zone at the coast to be a general feature of the zone, by supposing that differ-

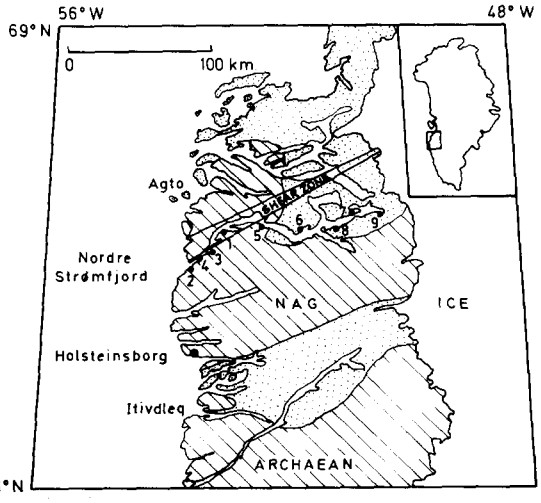


Fig. 1. Map of part of West Greenland showing sampling sites in Nordre Strømfjord (granulite areas are lined and amphibolite areas dotted).

ential uplift had occurred to the extent of about 10 km [1,2], so that a deeper crustal level is exposed at the mouth of Nordre Strømfjord than at its eastern end. This geological model has been confirmed by subsequent mapping in 1975 [3] and 1976. Movement of the crustal segment under consideration may have happened in two stages (Fig. 2): (1) uplift and erosion, following the Nagssugtoqidian metamorphism [4]; and (2) differential uplift, due to opening of the Labrador Sea.

Palaeomagnetism may be used to test this model. If magnetization occurred during the first phase of uplift and associated slow cooling [5,6], and if polar wander took place in this cooling period, it should be possible to discern an apparent polar wander path (APWP) through points of later and later magnetization, when moving from the ice towards the coast. This paper reports the result of such an experiment and of a further project performed at Holsteinsborg, which involved sampling up Kællingehætten in the expectation that a second APWP would be obtained which could be related to the previous path and thus provide another estimate of the postulated differential uplift in the fjord. One simplifying factor is that secular variation has probably been averaged out [5] because the magnetizations would have been acquired in a time much longer than the period of secular variation.

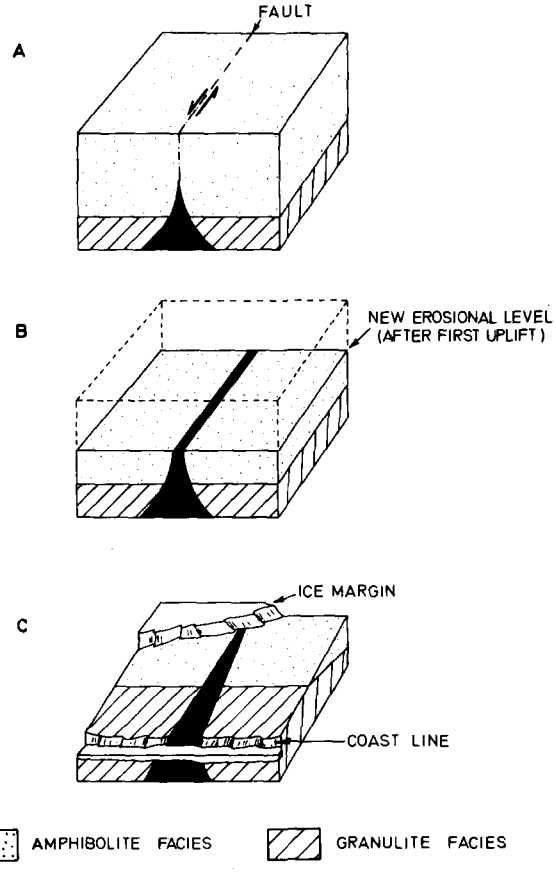


Fig. 2. Uplift model of the Nordre Strømfjord area. First uplift: A to B. Second uplift: B to C.

2. Field work

Nordre Strømfjord. Nine sites were sampled along the fjord (Fig. 1). In order to obtain precise palaeomagnetic pole positions, approximately 25 samples (8 cm long and 2.5 cm in diameter) were drilled at each site. They were orientated mainly by sun-compass [7] but in a few cases by Brunton compass in conjunction with topographic sighting.

Kællingehætten. This mountain (775 m) overlooking Holsteinsborg consists of granulite facies gneisses; it proved possible to take the equipment up to 623 m. Three other levels on the mountain were sampled. Sea-level sampling was done at the GGU house at Holsteinsborg, in the same band of granulite as was sampled near the top of the mountain (L. Davidson, personal communication, 1975). The heights of the sites (Table 1) were calculated from the readings of a Baromec altimeter. After the field work 2.5 cm lengths were cut from each sample to form standard specimens for magnetometer measurements.

3. Laboratory work and results

3.1. Measurements of magnetization

Intensities and directions of magnetization were measured on a PAR SM-2 spinner magnetometer (except for two weakly magnetized sites (q.v.)).

Since ancient components of magnetization are invariably overlain by secondary components, it is necessary to apply a magnetic cleaning procedure, usually partial alternating field (AF) or thermal demagnetization. The latter seemed preferable because it was more likely to reverse the natural process by which the magnetization had been acquired. Thermal demagnetization was applied in five steps to two Nordre Strømfjord sites and two sites from Kællingehætten to see whether it was appropriate to use this method upon the whole collection and, if so, which demagnetizing temperatures would be suitable. All specimens from these four sites were treated at (20), 300, 400, 500, 540, 560°C. The precision of grouping of directions is expressed by the precision parameter k (Fig. 3), whilst the semi-angle (in degrees) of the cone of 95% confidence about the mean direction is given by α_{95} [8]. Heating caused movement of the mean direction of magnetization at first but at 540°C it became almost stationary, except for NS9. It was therefore decided that all the rest of the rocks would be treated to this temperature and that the resulting direction of magnetization would be taken as the cleaned direction in each case. Previously all rocks would be given treatment at 500°C to monitor the direction at 540°C. The choice of these two demagnetizing temperatures has largely been borne out by the results for the other units. For example in NS2, NS4, NS3, H2, H3 and H4 the mean direction of magnetization moved 1°, 2°, 1°, 2°, 2°, 1°, respectively, between the two steps. Partial thermal demagnetization thus appears to be a useful method for isolating the stable direction of magnetization in these gneisses.

The magnetic susceptibilities of the specimens were measured [9] before and after each thermal treatment to check that the heating did not cause a chemical change in the minerals responsible for the magnetization. No such gross changes were found. Any changes amounted generally to less than 20% in susceptibility values which were probably due to changes in domain wall configurations in magnetite.

Two of the sites in Nordre Strømfjord were very weakly magnetized. However it proved possible to measure their directions of magnetization on the Digico spinner magnetometer [10]. Whereas the rocks in this study had typical values of intensity $\sim 10^{-4}$ emu/cm^3 after thermal demagnetization, the corresponding figures for NS7 and NS8 were $\sim 10^{-6}$ and $\sim 10^{-8}$, respectively.

The palaeomagnetic poles (Table 1) for the Nordre Strømfjord sites have been plotted in Fig. 6a (except NS9 whose pole lies near the geographical pole and does not have a stable end-point; this site is rejected). The pole for NS6 lies far from the main group but will be considered again; however at the present stage, the position of NS6 makes it difficult to see how Fig. 6a can be interpreted in terms of apparent polar wander. The palaeomagnetic poles from the five sites on Kællingehætten are plotted in Fig. 7a.

Experiment at higher temperatures. All sites had now been treated to temperatures a little lower than the Curie point of magnetite (578°C), in the belief that this was the mineral responsible for the magnetization. However, the possibility existed that haematite was also present. To test this, it was decided to subject site Hl to treatment at higher temperatures. The remanence fell sharply at the Curie point of magnetite (Fig. 3) but did not vanish; it persisted up to the haematite Curie point (680°C) with an intensity of about 1% of the original remanence. Inevitably when the remanence fell to such small values the grouping of directions also fell sharply but the mean direction for the site remained significant at the 95% confidence level, up to the Curie point of haematite (and apparently slightly above, but temperature differences within the furnace may account for this) and was not significantly different from that at lower temperatures. When magnetite and haematite are present simultaneously in rocks which have been very slowly cooled, it is to be expected that the haematite will display an older palaeomagnetic direction than the magnetite [5]. This experiment therefore suggests the possibility that Hl acquired its magnetizations during a phase of slow polar wander.

3.2. Measurements of magnetic susceptibility ellipsoids

The results described above, especially the Nordre Strømfjord poles, did not accord with the expectations of this study; it was therefore particularly important to be sure that the directions of magnetization were a correct record of the ancient magnetic field. The most obvious source of error lay in aniso-

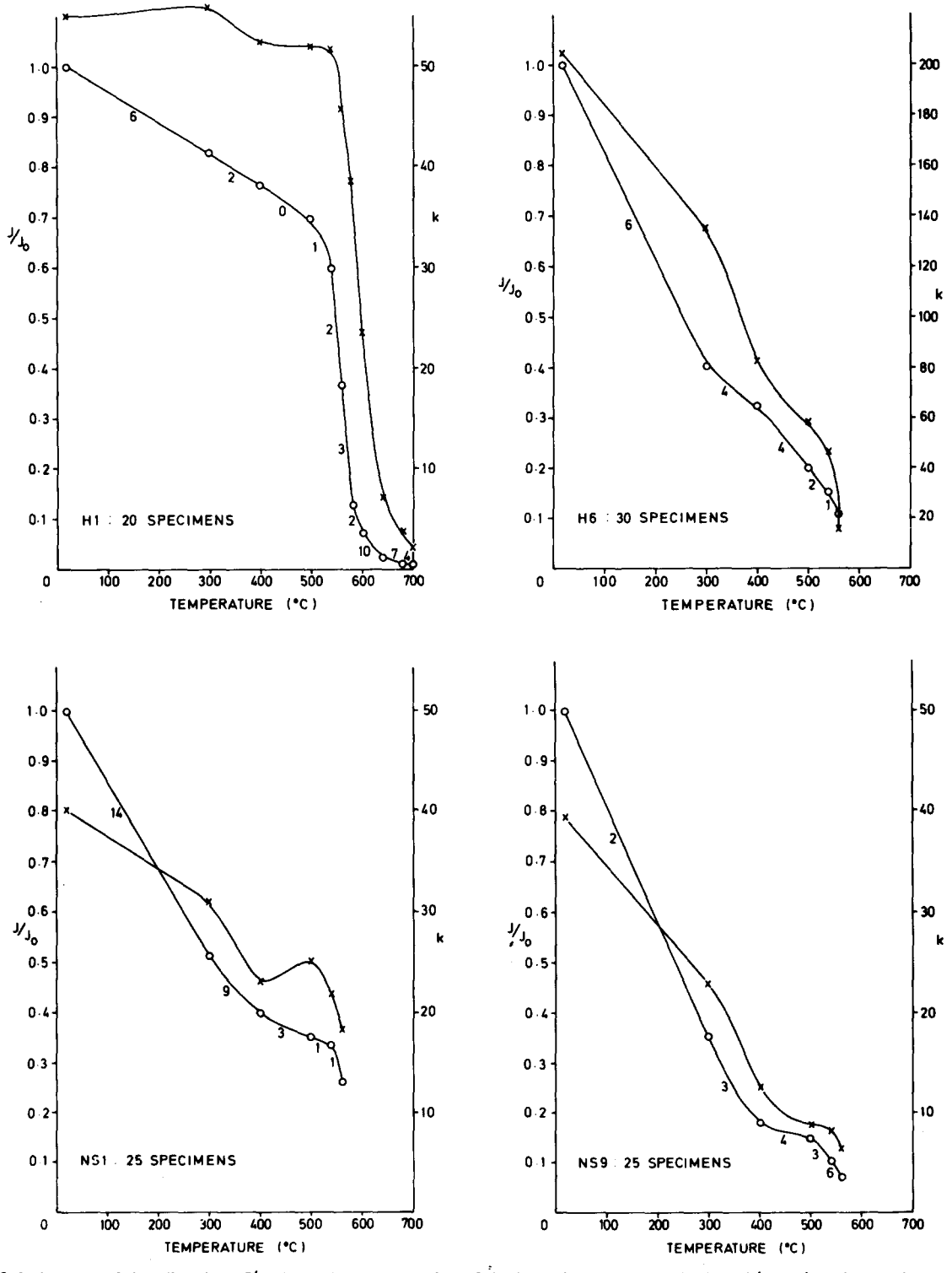


Fig. 3. Behaviour of the pilot sites. J/J_0 (○) is the average value of the intensity relative to the initial intensity; the numbers between the points are the angles in degrees turned by the mean direction. k (×) is the precision parameter.

tropy of the magnetic susceptibility which causes magnetization to lie at an angle to the magnetizing field. As fabric anisotropy of gneisses in the shear zone is higher than average this point seemed to be of particular interest. Accordingly it was decided to investigate this factor by measuring the directions and values of the principal components of the susceptibility ellipsoid ($\chi_{max}, \chi_{int}, \chi_{min}$) for each specimen. The measurements have been made on the Complete Results Anisotropy Delineator (CRAD), a development of the spinner magnetometer [10]. The three values are based upon a previous determination of the bulk susceptibility, χ. Susceptibility directions are double-ended; the CRAD yields six clusters of directions from a rock site but these may be reduced to three clusters by favouring (say) downward inclinations. The ratio χ_{max}/χ_{min} for each specimen is known as the anisotropy. Average anisotropies are listed in Table 1 and directions of the susceptibility axes at a typical site are shown in Fig. 4a.

3.3. Palaeomagnetic correction for anisotropy of susceptibility

An error of 3° is sometimes quoted as the maximum error arising from anisotropy which should be allowed. In that case it can easily be shown that the maximum permissible anisotropy is 1.1. However, in Nordre Strømfjord anisotropies are sometimes more than 10.0. Approximate directions of the ancient magnetic field may be found by picking out the less anisotropic specimens. If the anisotropy is <1.5 it can be shown that the deflection of the remanence away from the field direction is $<11.5°$. Nineteen specimens from Nordre Strømfjord and thirty-five from the mountain have anisotropies with such values and so, setting aside for the moment the question of polar wander between sites, estimates are obtained for the directions of the ancient magnetic field (Table 2). In Nordre Strømfjord, by a coincidence, this estimate of the field direction falls near the maximum-intermediate susceptibility planes for NS2 (Fig. 4a) and for NS1 and NS5. For this reason the angles between the ancient magnetic field and the magnetization are smaller than might have been anticipated.

A method for correcting for anisotropy of the magnetization has been based upon the following hypothetical approach. The intensity of thermoremanent magnetization (TRM), resulting from a field H, is given by $J_{TRM} = Q\chi H$ where χH is the intensity induced at room temperature by the same field and Q is the Koenigsberger ratio [11]. This equation suggests that the intensities of TRM resulting from the field H resolved into components H_1, H_2, H_3 along the directions of maximum, intermediate and minimum susceptibilities might have the forms:

$$J_1 \propto \chi_{max} H_1$$

$$J_1 \propto \chi_{int} H_2$$

$$J_3 \propto \chi_{min} H_3$$

As a first approximation, assume that the constant of proportionality is the same for each relation. The three components of J can then be combined to give the direction of TRM.

These assumptions have been tested experimentally. Ten specimens were selected, six from the fjord and four from the mountain. They were placed on their bases in a furnace and caused to acquire thermoremanent magnetization by cooling them from 600°C to 300°C in the vertical component of the Earth's field and then to room temperature in zero field. (This temperature range was used because a previous experiment from 300°C to room temperature was unsatisfactory.) Subsequently the directions of magnetization were measured and compared with the directions predicted by our hypothesis (Fig. 5a). Clearly the agreement is quite good.

This experiment suggests a method of correcting remanence directions by carrying out the calculation in reverse. This was done for the ten directions determined in the experiment. In each case the corrected direction lay nearer the vertical. On average the uncorrected directions were 14° away from the vertical whereas the corrected directions lay 6° from it whilst the precision parameter improved from 32 to 192 (Fig. 5b). Although the agreement with the true field direction is imperfect, our correction method appears to offer a useful approximation to the ancient field direction in metamorphic rocks. Accordingly it was applied to every specimen in the present collection and mean directions for each site calculated (Table 2). However in NS3, χ_{min} is effectively zero so that the ancient field is indeterminate for this site.

TABLE 1

Palaeomagnetic results after treatment at 540°C (uncorrected) and average anisotropies for all sites; heights of sites on the mountain

Rock unit	No. of specimens	Magnetization				Palaeomagnetic pole				Average anisotropies	Height (m)
		inc.	dec.	k	α_{95}	lat. (°N)	long. (°E)	dp	dm		
NS2	26	44.6	234.4	61	3.7	12.0	258.1	2.9	4.6	8.9	
NS4	30	63.2	264.0	15	7.1	38.4	242.2	8.8	11.1	36.6	
NS3	25	62.1	213.6	32	5.2	23.8	280.7	6.2	8.0	~100	
NS1	25	50.1	229.3	22	6.3	15.2	264.6	5.7	8.5	3.4	
NS5	25	63.4	229.9	24	6.1	28.7	269.5	7.6	9.6	1.5	
NS6	26	67.5	105.0	13	8.2	40.5	2.6	11.3	13.6	3.4	
NS8	20	59.9	266.0	3	26.2	35.8	240.4	29.9	39.6	1.5	
NS7	25	60.6	234.3	3	20.4	26.7	266.4	23.6	31.0	2.3	
NS9	25	77.0	28.5	8	10.8	78.5	38.9	18.7	20.1	2.0	
H1	20	72.7	213.4	52	4.6	37.4	284.9	7.2	8.1	1.3	623
H2	25	69.7	225.8	42	4.5	35.2	275.0	6.7	7.8	2.5	497
H3	25	74.3	183.6	7	12.2	37.6	304.2	20.1	22.2	2.9	407
H4	32	77.3	251.3	13	7.3	51.9	267.3	12.7	13.6	2.9	323
H6 *	30	73.9	270.3	47	3.9	52.9	250.3	6.3	7.0	2.3	5

dp and *dm* express the errors in the pole at the 95% confidence level. They define the semi-axes of the oval of confidence, *dp* in the ancient co-latitude direction (measured from the sampling site) and *dm* at right angles to it.
* A site H5 at 167 metres had to be abandoned.

TABLE 2

Palaeomagnetic results corrected for anisotropy of susceptibility, and between site statistics

Rock unit	Magnetization				Palaeomagnetic pole					Magnetization				Palaeomagnetic pole			
	inc.	dec.	k	α_{95}	lat. (°N)	long. (°E)	dp	dm		inc.	dec.	k	α_{95}	lat. (°N)	long. (°E)	dp	dm
	Palaeomagnetic results									*Between site statistics*							
NS2	52.1	222.4	9	9.8	15.1	270.3	9.2	13.5		(a) before correction; (b) after correction; (c) estimates of the ancient magnetic field directions, based on specimens with anisotropies <1.5							
NS4	66.8	218.4	7	10.3	30.4	278.6	14.1	17.1									
NS3	indeterminate																
NS1	45.8	202.9	21	6.5	6.5	286.6	5.3	8.3	(a) *	64.2	236.3	14	16.7	31.1	264.9	21.2	26.6
NS5	62.3	232.4	25	6.0	28.1	267.0	7.2	9.3	(b) *	64.2	220.8	32	10.9	27.7	276.6	13.8	17.3
NS6	80.4	177.9	13	8.1	49.1	309.4	14.9	15.5	(c)	56.8	241.0	18	8.1				
NS8	65.3	255.7	3	24.0	38.2	252.4	31.6	39.0									
NS7	68.0	220.3	3	19.5	32.7	280.4	27.5	32.8									
NS9	rejected																
H1	69.3	198.7	52	4.6	30.6	293.4	6.7	7.8	(a)	75.5	227.8	77	8.8	44.2	278.1	14.8	16.1
H2	57.7	186.2	52	4.1	15.4	301.3	4.4	6.0	(b)	65.9	188.2	79	8.7	25.3	300.4	11.5	14.1
H3	56.5	176.8	11	9.4	14.0	309.0	9.8	13.6	(c)	70.9	209.4	36	4.1				
H4	69.6	200.2	11	8.2	31.3	292.5	12.1	14.1									
H6	75.1	185.8	36	4.4	39.0	302.8	7.4	8.1									

* Nordre Strømfjord statistics are based on 7 sites, i.e. all except NS3 and NS9

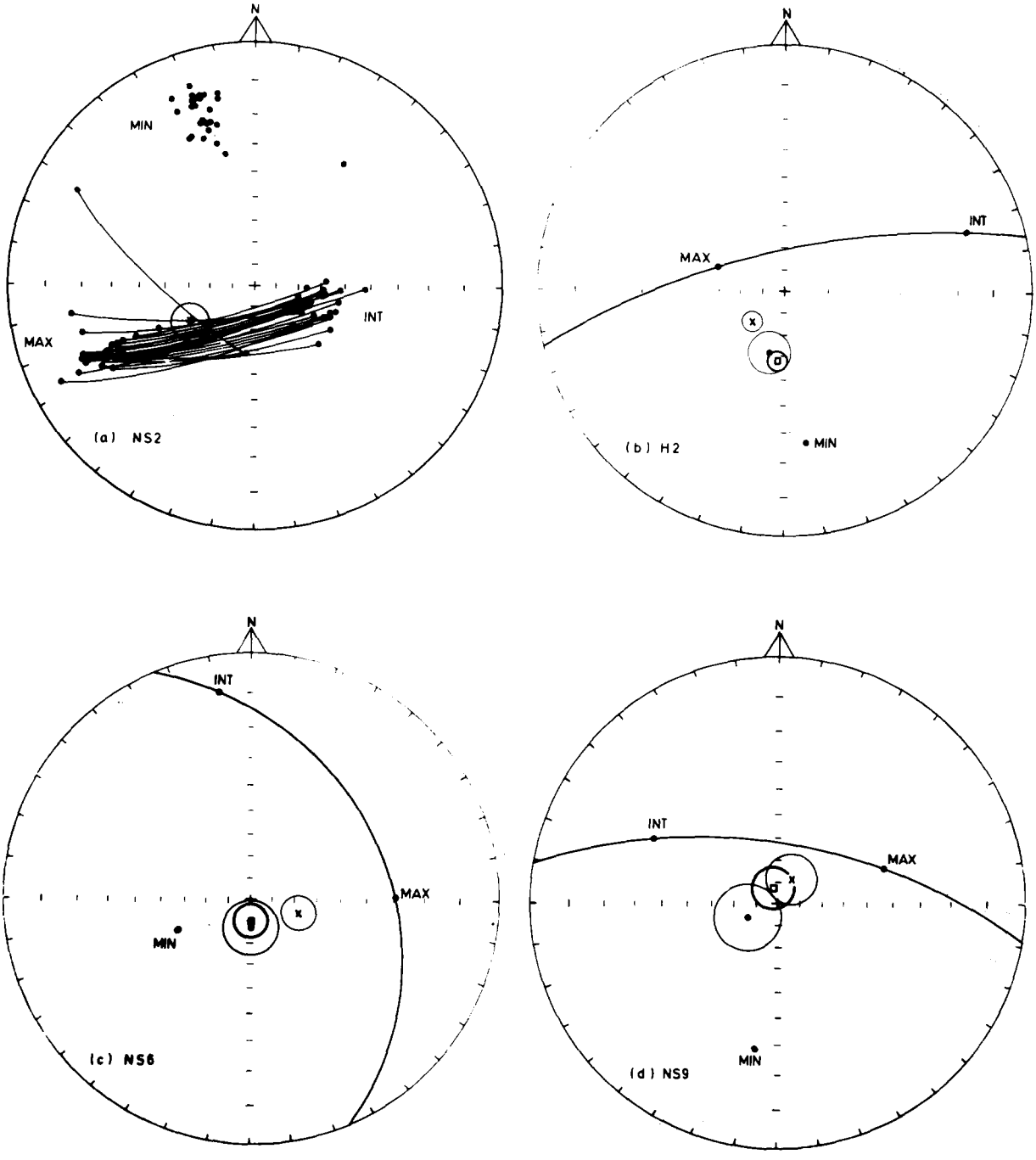


Fig. 4. MAX, INT, MIN represent the directions of the susceptibility components, referring to individual specimens in (a) and means in (b), (c), (d). (a) In NS2, the estimate of the true direction of the ancient field for Nordre Strømfjord (*) falls near the MAX/INT planes for individual specimens. NS1 and NS5 display similar behaviour. (b), (c), (d) illustrate the correction of the apparent mean direction of magnetization (X) to the original field direction (□). The diagrams also show the "statistical trial" in which the filled circle (●) is the corrected direction for the 7 least anisotropic specimens in a site. The circles denote limits of 95% confidence about mean directions. These diagrams are equal-angle projections.

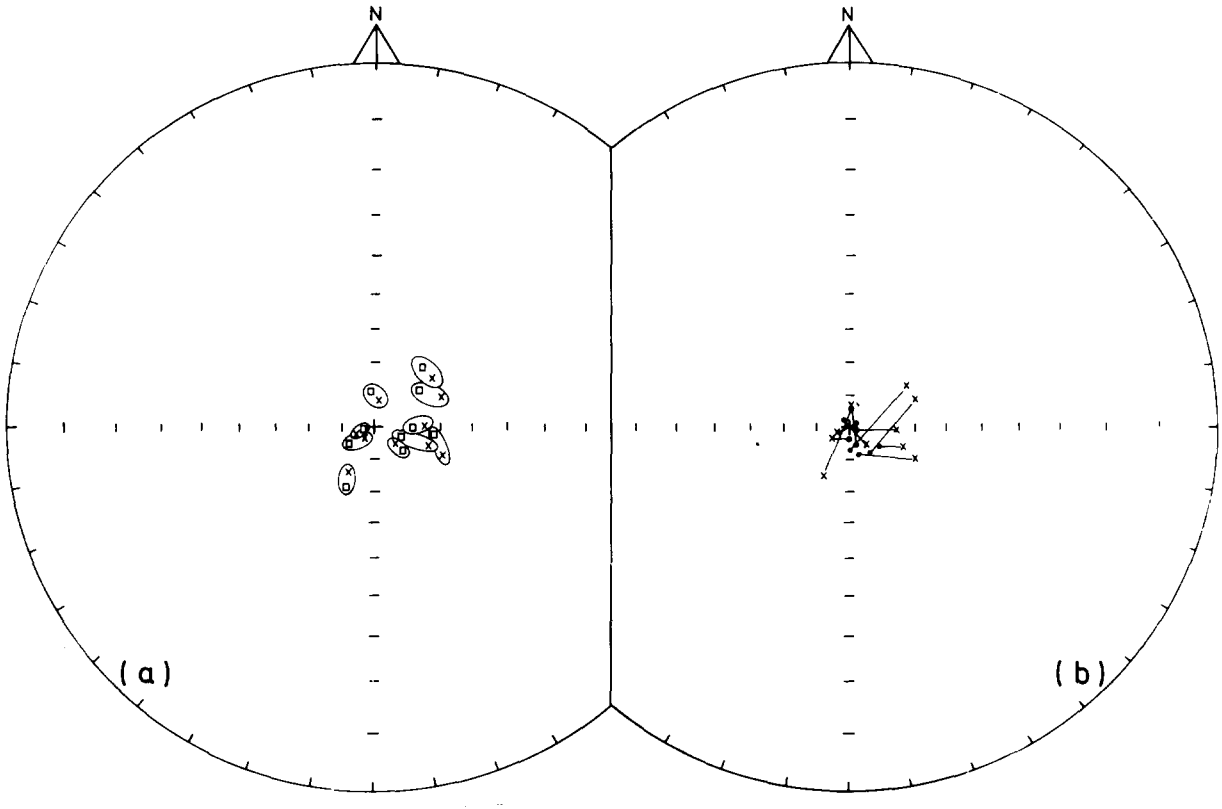


Fig. 5. Experimental test of the anisotropy correction method. The field direction was vertically down the 10 specimens, corresponding to the centre of the diagrams; azimuths are relative to their fiducial marks. (a) The cross (×) shows the measured direction of artificial TRM and the open square (□) the predicted direction for each specimen. (b) Relation of measured directions (×) to "corrected" directions (•).

(The exclusion of this site is regarded as an important negative outcome of the susceptibility work because the uncorrected result (Table 1) was apparently valid.)

In metamorphic rocks generally, the directions of the axes of the susceptibility ellipsoids are governed by the rock fabric [12]; in particular this is true of certain West Greenland gneisses [13]. These directions were obviously established before the magnetizations.

The experimental test of the correction method has been supplemented by a statistical trial. The same mean corrected direction should result within a particular site whatever the values of the susceptibility components. This proposition has been tested by picking the seven least anisotropic specimens from each site and determining a corrected direction from these. In most cases the agreement with the corrected direction for the whole site is excellent (e.g. Fig. 4b, c). NS9 is exceptional (Fig. 4d), reinforcing the earlier decision to reject this site. The trial is inconclusive on NS4, on account of the very high anisotropies. (NS7 and NS8 have been omitted from the trial on account of the scattering at these sites.)

For Nordre Strømfjord the most important effect of the correction has been to group NS6 with the other sites (Fig. 6b.)

4. Discussion and conclusions

If the imprecise poles for NS7 and NS8 are omitted, NS6 may be assumed to be the oldest (corrected) palaeomagnetic pole from Nordre Strømfjord.

An APWP may then be drawn through the poles of the other valid sites in order along the fjord (Fig. 6b). For the mountain the site at the top is expected to have the oldest magnetization and so an APWP has been drawn from H1 through the intervening sites to H6 (Fig. 7b). These paths are more complicated than had been anticipated, making it difficult to draw simple conclusions satisfying the objectives of the survey.

One problem inherent in all such complex paths is to decide whether they really are APWPs or whether the points on the path merely reflect scattering about a mean pole position. In Fig. 6b adjacent palaeomagnetic poles in the sequence NS6, NS5, NS1, NS4, are well separated. We therefore favour an interpretation in terms of polar wander, whilst recognizing that an element of scattering, perhaps due to faulting [5], is likely to be present. This interpretation is consistent with the geological model but contributes no new information to it.

The complex character of these APWPs recalls the complexity noted elsewhere in Greenland for rocks magnetized between 1850 and 1600 Ma [14,15], and for the early Precambrian of North America [16–18]. These paths and our own APWPs are confined to a portion of the globe within (or just outside) the present area of North America. This is also the case for the palaeomagnetic poles for the Godthåb gneisses [19], the Kangâmiut dykes at Itivdleq [5,14] and gneisses at Angmagssalik in East Greenland [20], which lie very close to each other and to the mean

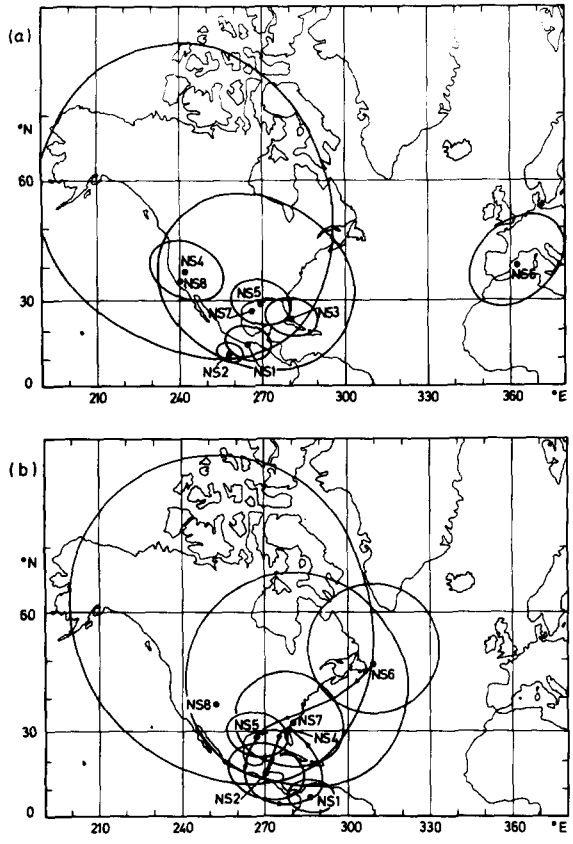


Fig. 6. Palaeomagnetic Poles for Nordre Strømfjord: (a) before correction; (b) after correction for anisotropy of susceptibility. (The poles in Figs. 6 and 7 are surrounded by ovals of 95% confidence; the maps are in Mercator's projection.)

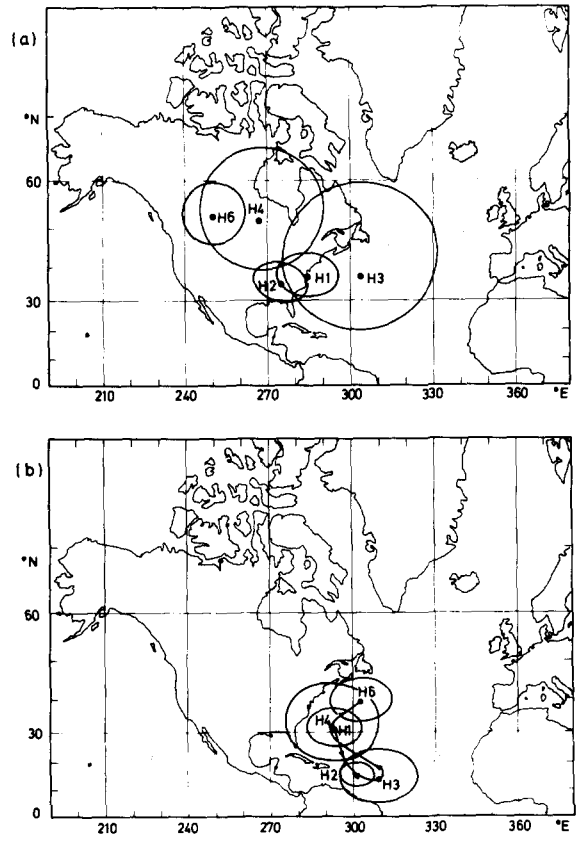


Fig. 7. Palaeomagnetic poles for Kællingehætten: (a) before correction; (b) after correction for anisotropy of susceptibility.

poles for Nordre Strømfjord and Kællingehætten. These observations invite the speculation that all these rocks were magnetized during times of slow polar wander; this suggestion is supported by our high temperature experiment on Hl.

For North America too Spall [21] recognized a quasi-static interval of polar wandering in the range 2060–1600 Ma; moreover rates of polar shift were generally low in the time range 1850–1650 Ma [22]. In spite of the low net rates of polar wander, the behaviour of the pole may have been complex in detail, making excursions away from a stable position and then returning, e.g. our nearly closed loops H1, H2, H3, H4 and NS5, NS1, NS4. Indeed it has been suggested [15] that apparent polar wander movements in Proterozoic times are most accurately described in terms of closed loops.

The assumption of magnetization during slow cooling requires a long episode of constant polarity of the Earth's field [23,5]. The magnetization of every specimen in this collection has positive inclination. This property, which is a striking feature of other early Precambrian rocks in Greenland, northwest Scotland [23–35] and North America, tends to reinforce our suggestion of polar stability. However, it is not yet possible to estimate the extent of this episode.

Unfortunately the low net rate of polar wander and the absence of reversals apparently render the palaeomagnetic method rather insensitive in these early times, at least in Greenland.

Acknowledgements

We thank GGU for their support during field work and for permission to publish. We thank L. Davidson and D. Craik for their noble effort in carrying the equipment up Kællingehætten. This work has been done during the tenure of a Leverhulme Fellowship by Beckmann; this and financial support from NERC are gratefully acknowledged. The maps have been drawn using Dr. R.L. Parker's program "Hypermap"; the incorporation of ovals of confidence is the work of Dr. M.L. Richards. The paper owes much to the referees and to colleagues at Newcastle. Beckmann wishes to dedicate this paper to the memory of Tony Richardson.

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